

***UNSW Law & Justice Research Series***

**The Missing Link: Legal and  
Regulatory Challenges in  
Interlinking Fast Payment  
Systems**

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[2026] *UNSWLRS* 4  
forthcoming NYU Journal of Law &  
Business

UNSW Law & Justice  
UNSW Sydney NSW 2052 Australia

**The Missing Link:  
Legal and Regulatory Challenges in Interlinking Fast Payment Systems**

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*This article presents the first comprehensive legal and policy framework to underpin the cross-border interlinking of Fast Payment Systems (F.P.S.). F.P.S. are innovative digital platforms that connect banks and non-bank payment service providers to exchange messages and settle transactions instantly. Although over 117 jurisdictions now operate F.P.S. and technical pilots are rapidly expanding, the legal architecture required for secure and interoperable arrangements remains insufficiently developed. The paper identifies key legal risks, including settlement finality, liability allocation, and AML/CFT compliance, and offers concrete solutions, including a publicly accessible Single Rulebook, prefunded accounts or a central counterparty to mitigate settlement risk, and enforceable dispute resolution mechanisms. Drawing on international best practices, it argues that legal harmonisation, transparency, and regional coordination are essential to achieving safe and efficient cross-border F.P.S. interlinking. The technological and regulatory framework proposed here directly supports the G20's 2027 target of reducing average remittance costs below three per cent and advancing global financial inclusion through faster, cheaper, and more accessible payment infrastructures.*

**Keywords:** fast payment systems, cross-border payments, remittances, settlement risk, retail payments

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## I INTRODUCTION

Fast payment systems<sup>1</sup> (F.P.S.) have become a cornerstone of modern domestic payment infrastructure, offering near-instantaneous fund transfers and expanding financial inclusion worldwide. F.P.S. are innovative digital platforms that link participating banks and non-bank payment service providers (P.S.P.s) to exchange payment messages and settle transactions in real time. As of June 2025, 117 countries have implemented a domestic F.P.S.,<sup>2</sup> and nearly 30% of the 45 monitored systems had at least one cross-border linkage, primarily within regional clusters such as Asia-Pacific and Europe.<sup>3</sup>

With growing remittance volumes, which surpassed \$800 billion globally in 2023 according to the World Bank, the inefficiencies in traditional cross-border payment channels have come under sharper scrutiny.<sup>4</sup> These inefficiencies include high transaction costs, delays, and limited transparency. For example, sending a \$200 remittance still costs an average of 6.18% globally and up to 11.48% through traditional banks.<sup>5</sup>

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<sup>1</sup> PRINCIPLES FOR FINANCIAL MARKET INFRASTRUCTURES defines a *payment system* as a “set of instruments, procedures and rules for the transfer of funds between or among participants”: BANK FOR INT’L SETTLEMENTS, COMM. ON PAYMENTS AND MARKET INFRASTRUCTURES, PRINCIPLES FOR FINANCIAL MARKET INFRASTRUCTURES 8 (Apr. 2012) [hereinafter *P.F.M.I.*].

<sup>2</sup> See Massimo Ferrari Minesso & Olga Triay Bagur, *Box 3: Geopolitics and Global Interlinking of Fast Payment Systems*, in EUR. CENT. BANK, THE INTERNATIONAL ROLE OF THE EURO 32 (June 2025), <https://www.ecb.europa.eu/pub/pdf/ire/ecb.ire202506.en.pdf>; Jon Frost et al., *Fast Payments: Design and Adoption*, B.I.S. Q. REV., MAR. 2024, at 31.

<sup>3</sup> See RSRV. BANK OF AUSTL., INTERLINKING FAST PAYMENT SYSTEMS FOR CROSS-BORDER PAYMENTS (Apr. 2024), <https://www.rba.gov.au/payments-and-infrastructure/pdf/report-interlinking-fast-payment-systems-for-cross-border-payments-april-2024.pdf> [hereinafter *Interlinking Fast Payment Systems*].

<sup>4</sup> See B.I.S. INNOVATION HUB, PROJECT RIALTO: IMPROVING INSTANT CROSS-BORDER PAYMENTS USING CENTRAL BANK MONEY SETTLEMENT (Feb. 2025), <https://www.B.I.S..org/publ/othp91.pdf>.

<sup>5</sup> *Id.*

Recognising these systemic challenges, the G20 has endorsed a Roadmap developed by the Financial Stability Board (F.S.B.) and the Committee on Payments and Market Infrastructures (C.P.M.I.) to make cross-border payments<sup>6</sup> faster, cheaper, more transparent, and more inclusive. A key milestone of this initiative is to lower the global average cost of remittances to below three per cent by 2027. Persistently high costs place a disproportionate burden on migrant workers and low-income households, underscoring why remittances are central not only to the G20's cross-border payments agenda but also to the United Nations Sustainable Development Goals (S.D.G.), particularly S.D.G. 10 (reducing inequality) and S.D.G. 16 (building inclusive institutions).

While various novel financial technologies have been proposed to achieve this, including stablecoins, wholesale central bank digital currencies (wCBDCs), and solutions like Wise, each has limitations. Stablecoins face regulatory and monetary sovereignty concerns; fintechs may lower costs but often take days to complete transfers in less-developed corridors; and wCBDC remain untested with unknown systemic risks.

By contrast, interlinking existing F.P.S.s offers a practical and near-term path toward achieving the G20's cost and speed targets for cross-border transfers. Rather than building entirely new infrastructure, interlinking arrangements are technology that leverage domestic systems already in place, improving liquidity management, compliance, and processing. These arrangements reduce reliance on correspondent banks and deliver faster, cheaper and more transparent remittances.

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<sup>6</sup> 'Cross-border payments' are defined in the *P.F.M.I.* as funds transfers for which the payer and the payee (i.e., the end users) are located in different jurisdictions. A cross-border payment may or may not involve currency conversion: *P.F.M.I.*, *supra* note 1.

Projects such as PayNow–PromptPay (Singapore–Thailand), UPI–PayNow (India–Singapore), T.I.P.S.–RIX (Europe–Sweden), and B.I.S.-led initiatives like Nexus and Rialto illustrate growing momentum. Over the next two years, 22 F.P.S. plan to establish their first or additional cross-border links, though many remain in early planning.<sup>7</sup>

Despite this progress, a major gap persists: few reports address the legal and regulatory foundations required to operationalise these linkages safely. This uncertainty underscores the urgent need for a clear, consistent legal framework to ensure settlement finality, liability, AML/CFT compliance, and governance are addressed from the outset.

This article addresses that gap by providing original legal and policy recommendations for structuring F.P.S. interlinking, including detailed guidance on a ‘Single Rulebook’ to ensure legal certainty, robust risk mitigation, and enforceability across jurisdictions. It also proposes using central counterparty clearing mechanisms (C.C.P.s) to manage cross-border liquidity and credit exposures while reducing systemic risk.<sup>8</sup>

Importantly, interlinking F.P.S.s is increasingly critical in the geopolitics of international finance, as it enables near-instant cross-border transactions while reducing reliance on dominant currencies and correspondent banking networks. By facilitating smoother

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<sup>7</sup> EMILIE FITZGERALD, ANAMARIA ILLES & THOMAS LAMMER, BANK FOR INT’L SETTLEMENTS, COMM. ON PAYMENTS AND MARKET INFRASTRUCTURES, STEADY AS WE GO: RESULTS OF THE 2023 C.P.M.I. CROSS-BORDER PAYMENTS MONITORING SURVEY (June, 2024), <https://www.B.I.S..org/C.P.M.I./publ/brief5.pdf> [hereinafter *Results of the 2023 C.P.M.I. Cross-Border Payments Monitoring Survey*].

<sup>8</sup> “Systemic risk: the risk that the inability of one or more participants to perform as expected will cause other participants to be unable to meet their obligations when due. In such circumstances, a variety of spillover effects are possible, and an FPS’s inability to complete inter-P.S.P. settlement could have significant adverse effects on the markets it serves and the broader economy,” BANK FOR INT’L SETTLEMENTS, COMM. ON PAYMENTS AND MARKET INFRASTRUCTURES, LINKING FAST PAYMENT SYSTEMS ACROSS BORDERS: GOVERNANCE AND OVERSIGHT 31 (Oct., 2024), <https://www.B.I.S..org/C.P.M.I./publ/d223.pdf> [hereinafter *Final Report to the G20*]. See SCOTT JAMES & LUCIA QUAGLIA, GEOECONOMICS AND FINANCIAL STATECRAFT IN THE EUROPEAN UNION: THE CASE OF CRITICAL MARKET INFRASTRUCTURES, 28 J. INT’L ECON. L. 490, 496 (2025).

settlement and enhancing financial resilience, interconnected systems can shift economic influence, promote regional financial integration, and provide states with strategic leverage in global payment infrastructures, making them a subtle but powerful tool in shaping economic power dynamics.

By focusing on legal architecture rather than technical design, this article aims to support policymakers, central banks, and payment system operators in building an interoperable, secure, and transparent global network for real-time payments.

## II FAST PAYMENT SYSTEMS (F.P.S.)

In the rapidly evolving fintech landscape, F.P.S. have become a cornerstone of retail payment infrastructure, enabling seamless, near-instant transactions. Originally designed for domestic use, F.P.S. now play a central role in enhancing efficiency, accessibility, and financial inclusion, forming a key part of payment system reforms in many jurisdictions.<sup>9</sup>

F.P.S. process payment messages and credit funds to the payee's account almost instantly,<sup>10</sup> operating continuously on a 24/7/365 basis. This distinguishes them from real-time gross settlement (R.T.G.S.) systems, which handle high-value wholesale payments<sup>11</sup> without transaction limits but typically within restricted operating hours. F.P.S. often set transaction

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<sup>9</sup> ALBERTO DI IORIO ET AL., BANK FOR INT'L SETTLEMENTS, COMM. ON PAYMENTS AND MARKET INFRASTRUCTURES, ACTA NON VERBA: INTERLINKING FAST PAYMENT SYSTEMS TO ENHANCE CROSS-BORDER PAYMENTS 3 (Feb. 2025), <https://www.B.I.S..org/C.P.M.I./publ/brief7.htm> [hereinafter *Interlinking Fast Payment Systems to Enhance Cross-Border Payments*].

<sup>10</sup> "An FPS is a retail payment system that allows for the transmission of the payment message and the final crediting of funds to the payee in real time or near real time. Final crediting in this context means that the payee has unconditional and irrevocable access to the funds, even if settlement among payment service providers (P.S.P.s) is deferred. FPS typically operate around the clock or very close to 24 hours a day, every day of the week throughout the year (24/7/365)," *Final Report to the G20*, *supra* note 8, at 3.

<sup>11</sup> The Financial Stability Board defines wholesale payments as transactions between financial institutions, typically exceeding US\$100,000, *Interlinking Fast Payment Systems*, *supra* note 3.

caps, such as the £1 million limit in the UK's Faster Payments Service, to manage risk, maintain stability, and efficiently process large volumes of low-value transactions.<sup>12</sup>

Operationally, F.P.S. are technology-enabled platforms that connect P.S.P.s to enable them to exchange payment messages and settle transactions in real time. To operate continuously outside central bank hours, P.S.P.s maintain prefunded settlement accounts, separate from reserve accounts<sup>13</sup> used for monetary policy. These accounts, held either at the central bank or within the F.P.S. infrastructure, are replenished during central bank hours from reserve balances.

Participation is generally limited to authorised financial institutions within the jurisdiction, subject to eligibility criteria and membership rules. Each F.P.S. operates under a set of operating rules and procedures, typically contained in a Rulebook, which functions as a contractual agreement between participants and the platform. F.P.S. platforms are typically overseen by the central bank or a designated authority to ensure operational stability, regulatory compliance, and effective risk management.

### **1. *The Settlement Process***

The defining feature of F.P.S. is that the payee gains near-instant access to funds once the payer initiates the transaction. At the customer level, the payee's account is credited almost immediately after the payer's account is debited, so settlement occurs in real time.<sup>14</sup> For

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<sup>12</sup> *Faster Payment System*, PAY.UK, <https://www.wearepay.uk/what-we-do/payment-systems/faster-payment-system/> (last visited Oct. 10, 2025) [hereinafter *Fast Payment System*].

<sup>13</sup> A reserve account is a deposit account held by a commercial bank at the central bank, used to settle interbank obligations and manage liquidity. Transfers between reserve accounts are typically executed through the Real-Time Gross Settlement (R.T.G.S.) system during its operating hours, allowing immediate and final settlement of high-value payments between participating institutions.

<sup>14</sup> A payer bank will always debit the payer's account after processing a payment instruction to ensure that there is no double-spending by the customer.

example, if Alice instructs her bank (the paying bank) to send \$100 to Bob, Bob's account is credited and the funds become available almost immediately after Alice's bank processes the payment.

At the P.S.P. level, however, interbank settlement between Bank A and Bank B may not occur instantly. The timing depends on the F.P.S. operator's model. F.P.S. generally adopt either real-time settlement,<sup>15</sup> where P.S.P. accounts are debited and credited in sync with the customer transaction, or deferred net settlement (D.N.S.), where the payee has immediate access to funds, but interbank settlement occurs later in scheduled batches. Both approaches ensure near-instant availability for the payee, even when P.S.P.-to-P.S.P. transfers are deferred.

## **2. Real-Time Settlement**

In a real-time F.P.S., each transaction is settled individually as it is initiated. When the payer instructs a transfer, the payer's P.S.P.'s account, which is held at the central bank or within the F.P.S., is debited, and the payee's P.S.P.'s account is credited simultaneously, ensuring immediate interbank settlement.<sup>16</sup>

For example, if Alice (Bank A) sends \$100 to Bob (Bank B), Bank A debits Alice's account and sends a payment message to the F.P.S. The F.P.S. instantly debits Bank A's

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<sup>15</sup> Real-time FPS should not be confused with R.T.G.S. systems. R.T.G.S. platforms handle large-value, wholesale payments between financial institutions, whereas FPS rails are designed for low-value, retail transfers between end users.

<sup>16</sup> Australia's New Payments Platform (N.P.P.) is an example of real-time settlement system, Alexandra Rush & Riaan Louw, *The New Payments Platform and Fast Settlement Service*, RSRV. BANK OF AUSTRALIA: BULLETIN (Sep. 20, 2018), <https://www.rba.gov.au/publications/bulletin/2018/sep/the-new-payments-platform-and-fast-settlement-service.html> [hereinafter *The New Payments Platform and Fast Settlement Service*].



prefunded account by \$100 and credits Bank B's prefunded account by the same amount. Bank B then immediately credits Bob's account, making the funds instantly available.

Because P.S.P.-to-P.S.P. settlement occurs in real time—often under a minute—there is no settlement or credit risk for the payee's bank. This prevents situations where an account is credited without receiving funds due to the payer bank's insolvency. While this model eliminates settlement risk by ensuring immediate finality, it demands high liquidity, as P.S.P.s must maintain substantial prefunded balances—a costly requirement for smaller institutions.

### ***3. Deferred Net Settlement***

In a D.N.S. system, transactions are accumulated over a defined period, and only the net amount<sup>17</sup> owed between P.S.P.s is settled at scheduled intervals. The payee's P.S.P. credits the customer's account immediately upon receiving the payment message, even though interbank settlement occurs later. At the close of each cycle, the F.P.S. calculates each P.S.P.'s net obligations and transfers only the net difference from prefunded accounts.<sup>18</sup>

This approach optimises liquidity and reduces the need for frequent large-value transfers but introduces potential settlement risk. If a net debtor P.S.P. defaults, eg, through insolvency, after the payee's P.S.P. has credited its customer's account, the latter may be unable

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<sup>17</sup> Netting is the process of offsetting multiple payment obligations between parties to determine a single net amount payable or receivable, thereby reducing the number and value of individual settlements required.

<sup>18</sup> The U.K.'s Faster Payments Service is an example of a D.N.S. system. Funds are credited to the payee's account in seconds, but interbank settlement occurs in netting cycles (typically three times daily) via accounts at the Bank of England: PAY.UK, FASTER PAYMENTS SERVICE PRINCIPLES: A GUIDE FOR PROSPECTIVE FPS PARTICIPANTS (Jan. 31, 2023), <https://www.wearepay.uk/wp-content/uploads/2023/02/Pay.UK-Faster-Payments-Service-Principles.pdf>.

to recover funds. In practice, this risk is minimised because P.S.P.s must maintain sufficient balances in prefunded accounts, ensuring coverage even if a debtor fails.

For example, Bank A sends \$100 from Alice to Bob at Bank B, while later Bank B sends \$50 from Craig to Dora at Bank A. At the end of the cycle, these obligations offset, leaving Bank A as the net debtor for \$50. The F.P.S. then debits \$50 from Bank A's prefunded account and credits Bank B's, completing the settlement.

By netting and offsetting transactions, D.N.S. requires only a single net transfer between P.S.P.s rather than settling every transaction individually. This greatly improves liquidity efficiency, as P.S.P.s need to prefund only enough to cover expected net obligations. However, it also creates exposure: if a net debtor P.S.P. fails to meet obligations at the cycle's close, net creditor P.S.P.s may face losses after crediting customer accounts without receiving the corresponding funds.

#### **4. *Settlement Finality***

In F.P.S., *operational* finality, which occurs when the payee's account and their P.S.P.'s account are credited, does not necessarily confer *legal* finality (also known as 'settlement finality'). In certain jurisdictions, insolvency provisions such as the 'zero-hour rule' may permit completed transactions to be unwound. Settlement finality is the legal moment when a payment becomes irrevocable and binding, even if a P.S.P. subsequently becomes insolvent. Without such protection, transactions may be reversed, posing both settlement and systemic risk. To safeguard payment integrity, the legal framework should explicitly protect settlement finality, and the F.P.S. Rulebook should clearly define when it is achieved.

#### **5. *The Zero-Hour Rule***

Many jurisdictions enacted legislation guaranteeing settlement finality to override the ‘zero-hour rule’, an insolvency law allowing administrators to retroactively void transactions made from midnight (00:00) on the day insolvency is declared, regardless of when proceedings begin.<sup>19</sup>

To illustrate the potential adverse effects of this, consider a scenario where Alice instructs Bank A at 14:00 to transfer \$100 to Bob. Bank A debits Alice’s account and credits Bank B’s account via the F.P.S., prompting Bank B to credit Bob’s account. If Bank A is declared insolvent at 15:00, the zero-hour rule could void all transactions since 00:00, leaving Bank B ‘out of pocket’ after crediting Bob and prompting it to recover funds from him. Similarly, if Bank B became insolvent at 16:00, its administrator could reverse Bob’s credit, creating uncertainty over payment finality.

In a D.N.S.-based F.P.S., similar risks arise at the close of the netting cycle. If Bank A became insolvent before settlement while owing Bank B, the administrator could block settlement, leaving Bank B exposed for funds already credited to Bob’s account.

Thus, the zero-hour rule undermines certainty in jurisdictions where it applies, as insolvency law can override the F.P.S. Rulebook—even when the Rulebook defines settlement finality as the moment the payee’s account is credited. Without specific settlement finality

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<sup>19</sup> For example, Australia retains a zero-hour rule in the *Bankruptcy Act 1966* (Cth) s 57A (Austl.): ‘Where, after the commencement of this section, a person becomes a bankrupt by virtue of the presentation of a debtor’s petition, the person shall, for the purposes of this Act, be deemed to become a bankrupt at the first instant of the day on which the petition is accepted by the Official Receiver.’ (emphasis added). A similar rule exists in the Netherlands: *Faillissementswet*, *Wet van 30 september 1893*, art 23 (Neth.), <http://www.dutchcivillaw.com/bankruptcyact.htm> [<https://www.viajuridica.nl/informatiesoorten/wet-en-regelgeving/faillissementswet>].

legislation, participants face credit and settlement risk, and legal uncertainty over when funds are truly irrevocable.

Such uncertainty undermines trust in F.P.S., as recipients like Bob may face retroactive reversals after spending the funds, eg, paying a supplier with the \$100, disrupting subsequent transactions and eroding confidence in the banking system. This instability can drive users toward less efficient alternatives, such as cash. Settlement finality legislation removes this risk by ensuring that once finality is reached, payments cannot be unwound, thereby safeguarding confidence and the smooth functioning of payment systems.

## **6. *Settlement Finality Legislation***

To guarantee legal settlement finality, many jurisdictions have enacted legislation to ensure that once a transaction is deemed final under Financial Market Infrastructures (FMI) rules, it becomes legally binding and cannot be reversed, even if a participant becomes insolvent. Such legislation exempts (carves-out) FMIs from the zero-hour rule and other corporate insolvency provisions.

In the European Union, the *Settlement Finality Directive* (SFD)<sup>20</sup> provides that “transfer orders and netting shall be legally enforceable and binding on third parties, even in the event of insolvency proceedings against a participant, provided that transfer orders were *entered* into a system before the moment of opening of such insolvency proceedings.”<sup>21</sup>

In the United States, article 4A of the Uniform Commercial Code (U.C.C.) governs funds transfers, with § 4A-406(a) stating that “a payment order is not revocable by the sender

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<sup>20</sup> Council Directive 98/26, art. 3, 1998 O.J. (L 166) 45 [hereinafter *Directive 98/26/EC*].

<sup>21</sup> (Emphasis added).

after it is *accepted* by the beneficiary's bank... the acceptance of the payment order by the beneficiary's bank creates an obligation of the sender's bank to pay the beneficiary's bank."<sup>22</sup>

Similarly, in Australia, section 14 of the *Payment Systems and Netting Act 1998* (PSNA)<sup>23</sup> provides that 'a payment or transfer *made* through an approved R.T.G.S. system or an approved netting arrangement is not to be voided, reversed, or repayable in the event of the external administration of a participant'.<sup>24</sup>

Across these jurisdictions, the legal framework ensures that once a payment reaches finality as defined in legislation or system Rulebooks, it is unconditional and irrevocable. This protects payees from the zero-hour rule and reinforces confidence in the payment system. Thus, when Bob receives \$100 from Alice, he can use the funds immediately without fear of reversal, even if a P.S.P. later fails.

## **7. Settlement Finality**

For F.P.S. operators, legal certainty requires defining settlement finality in the Rulebook as the moment the payee's account is credited, not when a payment instruction is merely submitted. Legal finality should be tied to an objective event—the actual crediting of funds—rather than the payee's expectation of payment. This ensures that only fully executed transfers are protected from reversal in insolvency proceedings, whether inter-P.S.P. settlement occurs in real time or on a D.N.S. basis.

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<sup>22</sup> U.C.C. § 4A-406(a) (UNIF. L. COMM'N 1977) (emphasis added).

<sup>23</sup> *Payment Systems and Netting Act 1998* (Cth) s 14 (Austl.).

<sup>24</sup> In Australia, 'The FSS enables final and irrevocable settlement of individual NPP transactions in real time... with a median settlement-processing time since launch of around 50 milliseconds,' *The New Payments Platform and Fast Settlement Service*, *supra* note 16.

This approach balances speed with security, making F.P.S. a dependable mechanism for everyday payments. It aligns with frameworks in the EU, US, and Australia, where finality occurs only when a payment order has been entered, accepted, and executed by the payee's P.S.P. In F.P.S., this point is reached when the payee's bank credits the account, which is an objective and verifiable point confirming the transfer has been completed by both the F.P.S. system and the payee's P.S.P.

## **8. *Knowledge of Insolvency***

If a P.S.P. becomes insolvent but fails to notify the F.P.S. operator, a key question arises: are payments processed in the interim still final and enforceable? Under Article 3(1) of the EU SFD, transfer orders entered into a designated system before insolvency are binding and irrevocable. Even orders entered afterward remain enforceable if the system neither knew nor could reasonably have been expected to know of the insolvency.

To ensure legal certainty, Rulebooks should require P.S.P.s to notify the operator immediately upon commencement of insolvency proceedings and should confirm that payments remain final until actual notification is received; constructive knowledge or suspicion should not suffice. This clear boundary protects system integrity and prevents retroactive clawbacks, which could otherwise undermine confidence in F.P.S..<sup>25</sup>

## **9. *Erroneous transfers***

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<sup>25</sup> If the operator receives notice of insolvency but continues processing payments, liability should shift to the operator. The Rulebook should make the operator contractually responsible for guaranteeing finality post-notification and compensating affected parties for losses. This explicit allocation of risk maintains settlement finality as robust, transparent, and enforceable, even during insolvency events.

An effective F.P.S. Rulebook must clearly define how to manage erroneous transfers, where a payer instructs their P.S.P. to transfer funds to the wrong account. In existing systems like the EU's TARGET Instant Payment Settlement (T.I.P.S.) and the UK's Faster Payments Service, the recipient may retain the funds until notified of the error. The payer must contact their P.S.P., which coordinates with the payee's P.S.P. to inform the recipient, who is expected to return the funds since they have no legal claim to them.

Because settlement finality prevents automatic reversal, the payee's bank cannot unilaterally recover the funds without the recipient's consent. If the recipient refuses, the payer can pursue legal action, often based on unjust enrichment or restitution, though outcomes vary by jurisdiction.<sup>26</sup> However, if the funds are spent in good faith before notification, the payer may lack legal recourse.

### ***10. Unauthorised and Fraudulent Transfers***

Unauthorised transfers, such as those caused by account hacking without the payer's authorisation, require explicit protections in F.P.S. Rulebooks. Where the payer bears no fault (eg, a system vulnerability rather than negligence such as weak password management), the F.P.S. should provide for compensation. This approach aligns with frameworks like the EU's *Payment Services Directive 2* (PSD2),<sup>27</sup> which mandates refunds for unauthorised transactions unless the payer is grossly negligent, and with existing banking practices where institutions investigate and may reimburse victims.

Reversing an unauthorised transfer to a fraudster's account typically requires court or regulatory authorisation, but P.S.P.s should, subject to regulatory approval, be able to

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<sup>26</sup> This approach is supported by *Directive 98/26/EC*, *supra* note 20, at recital 13.

<sup>27</sup> Parliament and Council Directive 2015/2366, arts. 69, 73, 2015 O.J. (L 337) 35.

temporarily freeze the recipient's account during investigation. This helps prevent further dissipation of funds and supports recovery efforts.

### **11. *Enhancing Security***

To minimise erroneous transfers, F.P.S. Rulebooks should require preventive controls such as two-factor authentication (2FA) and payee verification before funds are released. For example, the UK's *Confirmation of Payee* service flags name mismatches, while T.I.P.S. requires Payment Service User verification,<sup>28</sup> obliging P.S.P.s to confirm the payee's identity prior to execution. Real-time transaction alerts further help payers detect and report mistakes promptly.

For unauthorised transfers, enhanced authentication measures, such as 2FA and biometrics, strengthen protection against hacking. Rulebooks should also mandate robust cybersecurity standards for P.S.P.s and require them to compensate payers for losses resulting from system breaches where the payer is not negligent. These measures foster a safer, more resilient payment ecosystem and reinforce public trust in F.P.S.

## **III INTERLINKING F.P.S. ARRANGEMENTS**

Interlinking arrangements<sup>29</sup> are contractual, technical, and operational frameworks that connect payment systems across jurisdictions, enabling participating P.S.P.s to exchange payments as if operating within a same system. In F.P.S., interlinking arrangements link domestic platforms to enable seamless, secure, and real-time cross-border transfers. Such arrangements enable P.S.P.s to make cross-border payments without belonging to the foreign F.P.S., meaning a P.S.P. in country A can send funds to a P.S.P. in country B without opening

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<sup>28</sup> Parliament and Council Regulation 2024/886, art 5c, 2024 O.J. (L).

<sup>29</sup> *Final Report to the G20*, *supra* note 8, at 3.



accounts or obtaining membership abroad.<sup>30</sup> Connectivity is achieved through software that routes payment instructions and related data between systems, reconciles differences in message formats, and does so cost-effectively.

Interoperability is enabled through standardised messaging protocols, such as ISO 20022,<sup>31</sup> and application programming interfaces (APIs).<sup>32</sup> A cross-border payment typically involves two legs: debiting the payer's P.S.P. in one jurisdiction and crediting the payee's P.S.P. in another, with a separate foreign exchange (FX) transaction if currency conversion is required.<sup>33</sup> These arrangements allow P.S.P.s in different jurisdictions to exchange payment messages and settle transactions within seconds, so the payee in Country B is credited almost immediately after the payer in Country A initiates the transfer.

As with domestic F.P.S., the defining feature of interlinking must be the near-instant crediting of the payee's account. While technical design, settlement models, and liquidity arrangements may differ, the user experience must remain seamless, predictable, and fast. This standard is essential across all interlinked corridors, as it underpins confidence, drives adoption, and ensures competitiveness with alternatives like stablecoins, which already enable near-

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<sup>30</sup> BANK FOR INT'L SETTLEMENTS, COMM. ON PAYMENTS AND MARKET INFRASTRUCTURES ET AL., EXPLORING MULTILATERAL PLATFORMS FOR CROSS-BORDER PAYMENTS (Jan. 2023), <https://www.B.I.S..org/C.P.M.I./publ/d213.pdf> [hereinafter *Exploring Multilateral Platforms*].

<sup>31</sup> ISO 20022 is a global standard for electronic data exchange between financial institutions, enabling richer payment data and greater interoperability. Already adopted by 73% of FPS—mainly in Europe and Asia-Pacific—its use is projected to reach 80% within five years. While 69% of FPS plan full migration for domestic payments, only 42% intend to do so for cross-border payments, with Europe expected to lead at 81% within five years, *Results of the 2023 C.P.M.I. Cross-Border Payments Monitoring Survey*, *supra* note 7. See also J. Benton Heath, *Constructing a Global Panopticon: Towards a Jurisprudence of Weaponized Interdependence*, 28 J. INT'L ECON. L. 505, 518 (2025).

<sup>32</sup> APIs enhance cross-border FPS by reducing manual intervention and enabling timely data exchange. Adoption is rising, with 93% of FPS planning to use APIs within five years, particularly in the Middle East and Africa. Current uses include submitting or amending payment instructions and sharing transaction data or notifications: *Results of the 2023 C.P.M.I. Cross-Border Payments Monitoring Survey*, *supra* note 7, at 13–14.

<sup>33</sup> *Final Report to the G20*, *supra* note 8, at 3. Francisco J. Quintana, *Dollar Dominance, De-dollarization, and International Law*, 28 J. INT'L ECON. L. 359, 366 (2025).

instant cross-border transfers. Even where inter-P.S.P. settlement follows a D.N.S. model, the payee's account should still be credited irrevocably and without delay. Guaranteeing this outcome is critical to the long-term success and scalability of F.P.S. interlinking.

## **1. Correspondent Banking Arrangements**

Cross-border payments have traditionally operated through correspondent banking, where banks maintain bilateral agreements and use intermediaries for payment services abroad. Funds do not physically move across borders; instead, a sequence of accounting entries ultimately credits the payee's account (minus fees).<sup>34</sup>

Because banks rarely have direct relationships with all foreign counterparties, payments often pass through a chain of correspondent banks.<sup>35</sup> These relationships rely on *nostro* and *vostro* accounts: a *nostro* account (our money in your bank) is a bank's account at a foreign institution, while a *vostro* account (your money in our bank) is the reciprocal account held for the foreign bank.

For example, if Alice in Hong Kong sends HK\$100 to Bob in the UK, her bank (Bank A) may route the payment via a US correspondent (Bank C) with accounts at both Bank A and Bob's bank (Bank B). Bank C debits Bank A's *nostro* account, credits Bank B's *vostro* account with the GBP equivalent, and instructs Bank B to credit Bob's account. Each intermediary adds costs, foreign exchange spreads, delays, and repeated AML/CFT and sanctions checks.

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<sup>34</sup> Douglas Arner et al., *Building Regional Payment Areas: The Single Rule Book Approach* (Bank for Int'l Settlements Working Paper, Paper No 1016, May 2022), <https://www.B.I.S..org/publ/work1016.pdf> [hereinafter *The Single Rule Book Approach*].

<sup>35</sup> *Final Report to the G20*, *supra* note 8.

## 2. *Benefits of Interlinking F.P.S.*

F.P.S. interlinking extends the speed, affordability, and convenience of domestic instant payments to cross-border transactions, directly addressing inefficiencies in the traditional correspondent banking model. The G20 and C.P.M.I. have prioritised it as a step in modernising global payments, emphasising its potential to deliver faster, cheaper, more transparent, and more inclusive cross-border services.<sup>36</sup> Given the widespread adoption of domestic F.P.S., linking them is seen as a practical near- to medium-term pathway toward safer and more efficient retail payments.<sup>37</sup>

In contrast, the current correspondent banking model often involves multiple intermediaries, driving up costs,<sup>38</sup> prolonging settlement—sometimes for days—and limiting access, especially for smaller banks and non-bank P.S.P.s<sup>39</sup> lacking established relationships. Opaque fees and exchange rates, coupled with heavy AML/CFT compliance burdens, further constrains remittances, SME trade, and financial inclusion, particularly in emerging markets. Each additional intermediary increases costs, delays, and credit risk, compounding inefficiencies.

F.P.S. interlinking shortens this chain by enabling P.S.P.s in different jurisdictions to transact directly through connected infrastructures. This reduces reliance on multiple bilateral arrangements, narrows FX spreads, improves fee and status transparency, and enables

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<sup>36</sup> *Interlinking Fast Payment Systems*, *supra* note 9, at 2.

<sup>37</sup> *Final Report to the G20*, *supra* note 8, at 2.

<sup>38</sup> “McKinsey (2016) reported that the average cost for a US bank to execute a cross-border payment via the correspondent banking network is 10 times higher than the cost of an average domestic payment,” *The Single Rule Book Approach*, *supra* note 34, at 11.

<sup>39</sup> Dominant correspondent banks will inevitably exploit their market power through higher than competitive fees, Ulrich Bindseil & George Pantelopoulos, *Towards the Holy Grail of Cross-Border Payments* 17 (Eur. Cent. Bank. Working Paper, Paper No 2022/2693, Aug. 2022), <http://dx.doi.org/10.2139/ssrn.4178863> [hereinafter *Towards the Holy Grail*].

settlement within seconds. By leveraging existing domestic infrastructures, banking networks, and K.Y.C. frameworks,<sup>40</sup> interlinking expands access to reliable payment services, particularly in underserved markets, and advances financial inclusion.

Some correspondent banking remains necessary, particularly for currency conversion and cross-border liquidity management, but in a more streamlined form.<sup>41</sup> Interlinking frameworks can route payment instructions through institutions that already hold reserve or F.P.S. accounts in multiple jurisdictions, reducing intermediaries while preserving access to foreign currencies and liquidity sources.

By easing infrastructural and regulatory bottlenecks, F.P.S. interlinking is not merely a technical upgrade but a strategic reform that advances global policy objectives, facilitates trade, supports economic growth, and expands access to the international financial system.

### **3. Challenges**

Creating a global F.P.S. interlinking arrangement involves complex legal, regulatory, governance, technical, and political challenges that must be addressed to build a secure, efficient, and widely adopted system. A primary obstacle is the significant differences between legal and regulatory frameworks across jurisdictions. Each country has its own rules on settlement finality, netting, consumer protection, AML/CFT compliance, and data privacy. These legal discrepancies complicate harmonisation and, if unaddressed, expose participants to heightened legal and financial risks.<sup>42</sup>

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<sup>40</sup> *Id.* at 46–47.

<sup>41</sup> *Interlinking Fast Payment Systems*, *supra* note 3.

<sup>42</sup> *Id.*

Regulatory alignment, particularly around AML/CFT obligations, adds further complexity. Varying national standards make it difficult to design a uniform compliance framework, and achieving mutual recognition or streamlined processes across borders requires substantial coordination. Implementation is further challenged by fragmented data protection regimes. Some jurisdictions, for instance, mandate data localisation or limit cross-border sharing, which are restrictions that directly hinder straight-through processing (S.T.P.) and shared screening utilities that could otherwise enhance compliance efficiency.<sup>43</sup>

Another critical challenge is governance. F.P.S. linkages require clear oversight and dispute resolution mechanisms, yet the absence of globally accepted standards means countries must negotiate bespoke arrangements to address their commercial, legal, regulatory, and operational concerns. Developing a Single Rulebook is both time-consuming and politically sensitive.

Political hurdles also hinder multilateral schemes. Securing jurisdictional partners is difficult given conflicting compliance expectations, divergent cybersecurity standards, and differing cultural or risk-tolerance norms. Many countries note that regulatory and supervisory reforms, often slow and politically fraught, are a prerequisite for progress.<sup>44</sup>

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<sup>43</sup> BANK FOR INT'L SETTLEMENTS, COMM. ON PAYMENTS AND MARKET INFRASTRUCTURES, LINKING FAST PAYMENT SYSTEMS ACROSS BORDERS: CONSIDERATIONS FOR GOVERNANCE AND OVERSIGHT 8 (Oct., 2023), <https://www.B.I.S..org/C.P.M.I./publ/d219.pdf> [hereinafter *Considerations for Governance and Oversight*]; *Interlinking Fast Payment Systems*, *supra* note 3.

<sup>44</sup> *Exploring Multilateral Platforms*, *supra* note 30.

Technical challenges also feature prominently. Although 70% of F.P.S. use ISO 20022, variations in interpretation and implementation require upgrades or full migrations before systems can participate in an interlinked arrangement.<sup>45</sup>

Finally, practical factors such as cost, public sector support, and business model feasibility remain barriers. According to the C.P.M.I., some jurisdictions cite weak government backing, limited demand, and unclear commercial models as significant barriers to multilateral interlinking platforms.<sup>46</sup> Without sustained political will and long-term investment from both governments and industry, these projects risk stagnation.

In this context, global institutions such as the Bank for International Settlements (B.I.S.), International Monetary Fund (I.M.F.), World Bank, and F.S.B. could accelerate progress by developing a unified Single Rulebook, offering technical assistance, and establishing minimum global standards for legal, regulatory, and operational interoperability. Their leadership is crucial to overcoming fragmentation and aligning the varied interests of participating jurisdictions. This article provides legal and regulatory solutions and recommendations to overcome these challenges.

#### IV TYPES OF INTERLINKING ARRANGEMENT

This section provides an overview of the different types of interlinking arrangements. “There is no one-size-fits-all approach for oversight of F.P.S. interlinking arrangements”,<sup>47</sup> as each payment corridor presents distinct legal, technical, and institutional challenges. The structure

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<sup>45</sup> *Interlinking Fast Payment Systems*, *supra* note 3; Fin. Stability Bd., G20 ROADMAP FOR ENHANCING CROSS-BORDER PAYMENTS: CONSOLIDATED PROGRESS REPORT FOR 2024 (Oct. 21, 2024) [hereinafter *Consolidated Progress Report for 2024*].

<sup>46</sup> *Exploring Multilateral Platforms*, *supra* note 30, at 5–25.

<sup>47</sup> *Final Report to the G20*, *supra* note 8, at 1.

must be tailored to the jurisdictions involved. F.P.S. interlinking may take the form of bilateral links, each with its own rules, or a common multilateral framework.<sup>48</sup>

A fundamental prerequisite is that both countries have operational F.P.S.s. This remains a major barrier in regions where F.P.S. infrastructure is underdeveloped, including parts of Africa, the Americas (such as Canada),<sup>49</sup> and the South Pacific (including New Zealand).<sup>50</sup> Without domestic F.P.S. capabilities, jurisdictions cannot participate in interlinking, underscoring the importance of broader global F.P.S. adoption for real-time cross-border payments.

Depending on the design, an interlinking arrangement may require a dedicated entity to operate the technical infrastructure, such as a central ‘hub’ or shared platform. It may also involve designated FX providers for currency conversion (where applicable) and settlement agents<sup>51</sup> to manage final settlement across systems.<sup>52</sup>

Interlinking fast-payment systems can be organised along four models.<sup>53</sup>

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<sup>48</sup> *Towards the Holy Grail*, *supra* note 39, at 42–43.

<sup>49</sup> Canada will implement a new FPS, the Real-Time Rail (RTR), with the technical build expected to be completed in Q3 2025, *The Real-Time Rail: Canada’s Fastest Payment System*, PAYMENTS CANADA, <https://www.payments.ca/systems-services/payment-systems/real-time-rail-payment-system> (last visited Oct. 10, 2025).

<sup>50</sup> *Interlinking Fast Payment Systems*, *supra* note 3. New Zealand is the “only country of the 38 in the OECD that has not implemented, or committed to implementing, a real-time payments network,” Ruairi O’Shea, *New Zealand Left Behind by Rest of the World on Payment Technology*, CONSUMER, (Jan. 23, 2025), <https://www.consumer.org.nz/articles/new-zealand-left-behind-by-rest-of-the-world-on-payment-technology>.

<sup>51</sup> *Interlinking Fast Payment Systems*, *supra* note 3, at 3.

<sup>52</sup> *Considerations for Governance and Overnight*, *supra* note 43, at 8.

<sup>53</sup> For a comprehensive discussion on the full details and strengths of each model, *see Exploring Multilateral Platforms*, *supra* note 30); *Final Report to the G20*, *supra* note 8.

1. In the *single access point* model, participants in one domestic payment system access a foreign system through a single entity that directly participates in the foreign system.
2. In the *bilateral link* model,<sup>54</sup> domestic participants can directly reach all participants in the foreign system via the link, not just through a single gateway.
3. In the *hub and spoke* model,<sup>55</sup> bilateral links between two or more payment systems (the spokes) are replaced by links to a common intermediary (the hub).
4. In the *common platform* model,<sup>56</sup> participants interact directly across borders on a single, integrated platform.<sup>57</sup>

Although similar in concept, the hub-and-spoke and common-platform models, referred here collectively as ‘multilateral platforms’, differ in two key respects. First, in a hub-and-spoke model, domestic F.P.S. (the spokes) and connected P.S.P.s follow the hub’s Single Rulebook for cross-border transactions but retain their own rulebooks for domestic payments. By contrast, a common platform applies one Single Rulebook to all connected F.P.S. and P.S.P.s for both domestic and cross-border transactions, eliminating separate domestic rulebooks.

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<sup>54</sup> For example, Directo a México (2005), connecting the US FedACH and Mexico’s R.T.G.S. system (SPEI); PayNow–PromptPay (Singapore–Thailand, 2021); PayNow–UPI (Singapore–India, 2023); and Hong Kong–PromptPay link (2023), *Considerations for Governance and Oversight*, *supra* note 43.

<sup>55</sup> For example, the Regional Payment and Settlement System (REPSS) of the Common Market for Eastern and Southern Africa, *id.*

<sup>56</sup> For example, the SADC-R.T.G.S., hosted by the South African Reserve Bank on its SAMOS infrastructure, *id.* at 24–25.

<sup>57</sup> “While all four interlinking models can achieve similar results, the common platform is, in a strict sense, not a form of interlinking since P.S.P.s participate in one and the same payment system,” *Exploring Multilateral Platforms*, *supra* note 30, at 5–25.



Second, hubs are typically designed only for cross-border payments, while common platforms support both domestic and cross-border activity. For the purposes of this article, all multilateral platforms are assumed to operate under a Single Rulebook for cross-border payments, and no further distinction is made between the two models.<sup>58</sup>

### **1. *Choice of Interlinking Arrangement***

Bilateral links remain the dominant F.P.S. interlinking model, representing 54% of current arrangements.<sup>59</sup> However, they face scalability limits:<sup>60</sup> each new connection requires separate technical integration, legal agreements, and operational processes, creating exponential complexity as networks grow. As the ECB observes, “a network of 20 countries would require 190 country-to-country links”,<sup>61</sup> highlighting the inefficiency of a purely bilateral approach. Reflecting this, the C.P.M.I. notes that “if an F.P.S. plans to connect with more than one F.P.S. over time, it may choose a hub-and-spoke model over a bilateral model”.<sup>62</sup>

Many bilateral links appear to be transitional ‘quick wins’, with at least half of F.P.S.s using them planning to migrate to multilateral models.<sup>63</sup> While multilateral platforms represent just eight per cent of current connections, their share is expected to rise to nine per cent within two years and 28% within five. By allowing each F.P.S. to connect once to a single platform rather than bilaterally to every other system, these models offer a more scalable and sustainable route to interoperability.

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<sup>58</sup> *Id.*

<sup>59</sup> *Results of the 2023 C.P.M.I. Cross-Border Payments Monitoring Survey*, *supra* note 7.

<sup>60</sup> *Interlinking Fast Payment Systems*, *supra* note 3.

<sup>61</sup> *Towards the Holy Grail*, *supra* note 39, at 43.

<sup>62</sup> Final Report to the G20, *supra* note 8, at 15.

<sup>63</sup> *Results of the 2023 C.P.M.I. Cross-Border Payments Monitoring Survey*, *supra* note 7.

Although establishing a multilateral platform usually requires creating a new legal entity and a Single Rulebook, which potentially extends implementation timelines, it streamlines future onboarding, fosters harmonisation, and improves cross-border efficiency. The B.I.S.-C.P.M.I.’s Nexus initiative illustrates this potential, by combining a unified technical standard, consistent processing rules, and competitive FX conversion to reduce fragmentation and enable seamless integration across jurisdictions.

## V GOVERNANCE AND OVERSIGHT

Governance and oversight are essential to ensuring the safety, efficiency, and trustworthiness of F.P.S. interlinking arrangements. *Governance* refers to the rules, structures, and decision-making processes that define the roles and responsibilities of participants, operators, and owners of a payment system. In contrast, *oversight* is a central bank function that monitors and assesses payment arrangements to ensure they meet public policy objectives, especially safety, efficiency, and systemic resilience.<sup>64</sup>

Together, these frameworks allocate legal and operational risks, ensure compliance, and maintain stability across jurisdictions. Without strong governance and oversight, interlinking arrangements risk legal uncertainty, reputational damage, or even financial losses—outcomes that would undermine user confidence and international cooperation.

### 1. *Governance*

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<sup>64</sup> BANK FOR INT’L SETTLEMENTS, COMM. ON PAYMENTS AND MARKET INFRASTRUCTURES, A GLOSSARY OF TERMS USED IN PAYMENTS AND SETTLEMENT SYSTEMS 37 (rev. ed. Mar. 2003), [https://www.B.I.S..org/C.P.M.I./glossary\\_030301.pdf](https://www.B.I.S..org/C.P.M.I./glossary_030301.pdf).

Multilateral models generally operate under their own governance framework for cross-border payments, codified in a Single Rulebook that defines the rights, obligations, and procedures of all participants. In addition, the platform remains subject to the laws and regulations of its jurisdiction of incorporation,<sup>65</sup> including corporate law and any applicable central bank oversight, depending on whether it is privately or publicly operated.

Governance structures may include a board of directors, shareholder or member assemblies, and consultative committees to ensure strategic oversight, accountability, and stakeholder representation. These roles are usually outlined in founding documents such as articles of association or incorporation. The governance framework must strike a balance between institutional robustness, which ensures legitimacy, accountability and compliance, and operational flexibility that supports timely decisions and avoids unnecessary procedural complexity.

## **2. Oversight**

Effective oversight of F.P.S. interlinking requires strong cross-border cooperation among regulators, since disruptions can affect participants and central banks across jurisdictions. While each domestic F.P.S. remains under its home central bank, the interlinking arrangement, particularly if constituted as a separate legal entity, may require formal designation as a payment system and oversight under the host jurisdiction's legal framework.<sup>66</sup>

The Continuous Linked Settlement (C.L.S.) system provides a precedent, operating under a cooperative oversight by 18 central banks and euro-area authorities.<sup>67</sup> As multilateral

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<sup>65</sup> *Considerations for Governance and Oversight*, *supra* note 43, at 12.

<sup>66</sup> *Final Report to the G20*, *supra* note 8, at 7.

<sup>67</sup> *Exploring Multilateral Platforms*, *supra* note 30, at 525.

platforms expand in scale and systemic importance, similar cooperative frameworks will be essential. However, legal and institutional barriers, such as the absence of procedures to recognise or supervise foreign FMIs, may complicate implementation.

Coordination among national overseers helps prevent legal conflicts, regulatory gaps, and fragmented supervision. As the G20 notes,<sup>68</sup> oversight of an interlinking arrangement may differ from oversight of its component F.P.S.s, making structured cooperation vital to safeguard legal certainty, systemic integrity, and user confidence.<sup>69</sup>

### **3. Ownership**

A well-designed F.P.S. interlinking arrangement requires a transparent ownership structure. While central banks own, operate, or act as settlement agents for over 90% of R.T.G.S. systems, F.P.S.s more often adopt mixed models: central banks typically serve as settlement agents, while private entities participate in ownership or operation in about half of cases.<sup>70</sup> Private ownership can operate effectively, as demonstrated by well-regulated FMIs such as LCH, one of the world's largest C.C.P.s, provided that a strong regulatory and supervisory framework is in place. Whether ownership, operation, and governance are centralised in one entity or distributed across several, public sector ownership—or at minimum strong public oversight—is best positioned to safeguard financial stability and mitigate systemic risk.<sup>71</sup>

### **4. Location of Link Operator**

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<sup>68</sup> *Final Report to the G20*, *supra* note 8, at 7.

<sup>69</sup> *Considerations for Governance and Oversight*, *supra* note 43.

<sup>70</sup> *Results of the 2023 C.P.M.I. Cross-Border Payments Monitoring Survey*, *supra* note 7.

<sup>71</sup> *Final Report to the G20*, *supra* note 8, at 7.

Choosing the home jurisdiction for an interlinking operator, particularly when structured as a separate legal entity, is a foundational legal and strategic decision. The jurisdiction of incorporation determines the applicable regulatory framework, including rules on consumer protection, AML/CFT compliance, data privacy, corporate governance, and oversight. It also determines the courts with jurisdiction, the recognition of settlement finality, and the enforceability of contracts, which are all essential for legal certainty in cross-border payments. The home jurisdiction should therefore possess an independent judiciary, robust commercial and insolvency laws, political and economic stability, and a credible regulator with experience overseeing systemically important FMIs.

Jurisdictions such as Belgium, Switzerland, Singapore, Germany, and Japan offer many of these qualities, with mature legal systems, top-tier financial regulation, and central roles in global payment networks. Ideally, the jurisdiction should also issue a widely used currency and host international standard-setters or key institutions—such as the ECB—to reinforce trust and legitimacy. The reputation of the home jurisdiction influences the arrangement’s ability to attract members, ensure legal compliance, and secure effective cross-border cooperation.

## **VI CONTRACTUAL ARRANGEMENTS AND RULEBOOK**

Arner et al. have proposed a global ‘Single Rule Book’ to guide regional payment integration, similar to the UCC in the US, setting minimum harmonised standards while allowing local adaptation. As the authors explain: “such a Single Rule Book could be instrumental in enhancing safety, efficiency, and integrity in cross-border payments” by establishing a harmonised legal and operational baseline for regional and global interoperability.<sup>72</sup> This research supports that proposal by offering detailed legal risk management provisions for a

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<sup>72</sup> *The Single Rule Book Approach*, *supra* note 34, at 4.

Single Rulebook, addressing settlement finality, liability allocation, data protection, and dispute resolution.

The Single Rulebook should serve both as a contractual framework for interlinking arrangements and as a practical guide aligned with international standards. As Baker McKenzie observes, past linkages have been hindered by “multiple parties with diverse interests and no international standards to use as a basis.”<sup>73</sup> Incorporating global best practices, such as the P.F.M.I. and FATF recommendations, would reduce legal fragmentation, build trust, and minimise operational uncertainty. Over time, the emergence of widely accepted international standards would further streamline linkages, strengthening efficiency, interoperability, and confidence across jurisdictions.

In F.P.S. interlinking, the Single Rulebook is the core legal and operational framework of a cross-border arrangement.<sup>74</sup> Legally, it acts as a binding multi-party contract<sup>75</sup> between the link operator and participating domestic F.P.S.s, defining roles, responsibilities, and obligations, and establishing a uniform legal basis across jurisdictions.<sup>76</sup>

To ensure enforceability, its terms must be “clear, easily understandable for the relevant audience, and consistent with the laws and regulations of those jurisdictions in which it operates.”<sup>77</sup> While the operator may also hold bilateral agreements with FX providers,

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<sup>73</sup> MONETARY AUTH. OF SING., BANK OF THAI. & BAKER MCKENZIE, PAYNOW-PROMPTPAY/PROMPTPAY-PAYNOW LINKAGE WHITE PAPER: A PLAYBOOK FOR ASEAN AND THE WORLD 17–24 (Apr. 2022), [https://abs.org.sg/docs/library/PayNow-PromptPay\\_Linkage\\_White\\_Paper.pdf?sfvrsn=9](https://abs.org.sg/docs/library/PayNow-PromptPay_Linkage_White_Paper.pdf?sfvrsn=9) [hereinafter *A Playbook for ASEAN and the World*].

<sup>74</sup> “Rulebooks for payment instruments are single sets of rules, practices, standards and implementation guidelines for the execution of such payments,” BANK FOR INT’L SETTLEMENTS, COMM. ON PAYMENTS AND MARKET INFRASTRUCTURES, SERVICE LEVEL AGREEMENTS FOR CROSS-BORDER PAYMENT ARRANGEMENTS: RECOMMENDATIONS AND KEY FEATURES 15 (Apr. 2024), <https://www.B.I.S..org/C.P.M.I./publ/d222.pdf> [hereinafter *Recommendations and Key Features*].

<sup>75</sup> *A Playbook for ASEAN and the World*, *supra* note 73, at 17–24.

<sup>76</sup> *Final Report to the G20*, *supra* note 8, at 8–9.

<sup>77</sup> *Recommendations and Key Features*, *supra* note 74, at 16.

settlement banks, or technical service suppliers, the Single Rulebook remains the master document governing the scheme. If an F.P.S. or P.S.P. breaches its provisions, the operator should have authority to suspend or terminate participation.

Crucially, the Single Rulebook must be contractually binding on every participant, with enforceability across every jurisdiction represented.<sup>78</sup> Domestic F.P.S.s may need to amend their own rulebooks or adopt supplementary provisions to align with the interlinking arrangement. Where full alignment is impossible, such as with differing settlement finality models, technical and legal mechanisms can still ensure interoperability.<sup>79</sup>

The Single Rulebook functions as both a contract and a service-level agreement,<sup>80</sup> harmonising<sup>81</sup> operations across diverse systems through common standards for messaging, settlement cycles, transaction limits, risk-sharing, AML/CFT compliance, data protection, and cyber resilience.<sup>82</sup> It must also provide enforcement tools—sanctions for non-compliance, dispute resolution processes, and alignment with P.F.M.I. standards where the arrangement is systemically important.<sup>83</sup>

Risk management is central to its design, with minimum standards for access, operational resilience, liquidity, and fraud prevention. Striking the right balance is critical: overly strict requirements may deter participation, while lax standards undermine trust.

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<sup>78</sup> Christian Chamorro-Courtland, *Counterparty Substitution in Central Counterparty (C.C.P.) Systems*, 26 BANKING & FIN. L. REV. 517 (2010).

<sup>79</sup> *Interlinking Fast Payment Systems*, *supra* note 3.

<sup>80</sup> *Id.*

<sup>81</sup> *Id.*

<sup>82</sup> *Considerations for Governance and Oversight*, *supra* note 43, at 8–9.

<sup>83</sup> *Recommendations and Key Features*, *supra* note 74, at 5–6.

Flexible tools, such as corridor-specific limits or the option to decline connections with certain F.P.S.s or P.S.P.s, help manage legal and operational risk while enabling scalability.<sup>84</sup>

Furthermore, amendments to the Single Rulebook should follow a clearly defined procedure set out within the Rulebook itself. This process must specify who may propose changes, which is typically the link operator, participating F.P.S.s, or a governance committee, and the thresholds for approval. Minor operational updates might be adopted by simple majority, while substantive amendments affecting settlement, risk management, or access criteria should require a qualified or supermajority vote to ensure broad consensus.

### **1. *Resolving Conflicts***

A domestic F.P.S. may join only if its Rulebook is consistent with, and does not conflict with, the Single Rulebook. The accession contract should state that the Single Rulebook takes precedence in any conflict, at least for cross-border transactions, as a binding condition of participation. Alignment must be maintained on an ongoing basis, with non-compliance subject to remedies such as fines, suspension, or expulsion.

Beyond rulebook alignment, a domestic F.P.S.'s laws must meet the minimum standards set by the Single Rulebook, such as settlement finality legislation and compliance with P.F.M.I. and FATF principles. If a jurisdiction's laws later fall out of alignment (eg, repealing settlement finality protections), the link operator should apply a graduated response: issue a notice or 'greylist' warning, allow a defined period for corrective action, and, if unresolved, suspend or terminate membership.

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<sup>84</sup> *Interlinking Fast Payment Systems*, *supra* note 3.



## **2. *Dispute Resolution***

Dispute resolution is essential in interlinking arrangements, where participants operate under different legal systems, languages, and regulations. The Single Rulebook should set clear procedures for handling erroneous transfers, fraud, failed settlements, misallocation of funds, security breaches, or liability disputes arising from cyberattacks or sanctions.

A multi-tiered approach is recommended. First, parties attempt direct resolution, supported by a secure case-management platform. Retail customers contact their P.S.P., which liaises with the foreign P.S.P. or escalates through its domestic F.P.S., which then engages the corresponding F.P.S. and P.S.P. abroad. While F.P.S. routing provides oversight, direct P.S.P.-to-P.S.P. communication may expedite resolution for large, regulated entities.

If informal resolution fails, disputes should proceed to arbitration, preferably initiated by the domestic F.P.S. (unless P.S.P.s are directly bound by the Single Rulebook). The Single Rulebook should designate a neutral forum, such as London, Singapore, or Geneva, offering procedural neutrality, confidentiality, and enforceability. Arbitration would apply only to F.P.S.s and P.S.P.s, not end-users.

To preserve system integrity in cases of default (eg, P.S.P. insolvency), loss-allocation mechanisms should backstop the process. Options include a default fund, collateral requirements, or, in advanced models, a C.C.P. to absorb losses and ensure settlement continuity. While disputes and defaults cannot be eliminated entirely, such safeguards contain losses, prevent contagion, and maintain trust, thereby enhancing the arrangement's resilience and scalability.

## **VII RISKS IN INTERLINKING ARRANGEMENTS**

Cross-border F.P.S. interlinking faces greater complexity than domestic systems due to fragmented legal, regulatory, and technological regimes, which heighten legal risk, AML/CFT inconsistencies, and uncertainty over settlement finality. Robust scheme rules and processing standards that enforce minimum risk requirements across jurisdictions can mitigate these challenges, ensuring integrity, interoperability, and the scalability of cross-border payments.<sup>85</sup>

## **1. Legal Risk**

Legal risk arises when the operation or enforcement of an interlinking arrangement is threatened by the unexpected application—or misapplication—of laws and regulations, or when legal outcomes are uncertain. In cross-border contexts, this risk is amplified by multiple legal systems with differing statutory protections, regulatory frameworks, and interpretations of key concepts.

For example, a payment deemed final in one jurisdiction may not be recognised in another, undermining settlement finality.<sup>86</sup> Similar inconsistencies can occur in netting enforceability, insolvency treatment, access criteria, and data protection, jeopardising both transaction certainty and the enforceability of obligations, particularly if domestic laws change after a system has joined.<sup>87</sup>

To mitigate these risks, the Single Rulebook should contain clear, consistent, and enforceable provisions across all jurisdictions, explicitly addressing divergences in settlement finality, data sharing, and AML/CFT obligations, and prescribing remedial measures.<sup>88</sup> Jurisdictions with conflicting laws conflict may need to amend legislation or adopt enabling

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<sup>85</sup> *Id.*

<sup>86</sup> *Exploring Multilateral Platforms*, *supra* note 30, at 3.

<sup>87</sup> *Final Report to the G20*, *supra* note 8, at 10.

<sup>88</sup> *Exploring Multilateral Platforms*, *supra* note 30.

provisions before joining. Continuous legal due diligence and engagement with domestic authorities are critical to detect and address changes affecting enforceability. Mutual recognition of the interlinking arrangement as a systemically important FMI can further enhance legal certainty.

Contractual safeguards, such as choice-of-law clauses, arbitration agreements, and flow-down obligations in domestic rulebooks, can also reduce ambiguity. While contracts alone cannot eliminate all conflicts, an internationally coordinated legal framework greatly strengthens the credibility, resilience, and enforceability of interlinking arrangements.

## **2. Operational Risk**

Operational risk is the potential for disruption or loss from failures in systems, processes, personnel, or external events such as cyberattacks or natural disasters. In F.P.S. interlinking, this risk is amplified by the complexity of coordinating real-time payments across multiple jurisdictions. Failures in the interlinking operator or any connected F.P.S., whether from outages, software bugs, human error, or cyber incidents, can delay or block cross-border transactions, creating ripple effects across the network. Since F.P.S.s operate 24/7/365, even brief disruptions can quickly erode public confidence.<sup>89</sup>

The cross-border nature of these arrangements increases the ‘attack surface’ for operational failures, requiring a robust, multi-layered resilience framework. This should include strong cybersecurity measures, system redundancies, distributed data centres, and real-

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<sup>89</sup> *Final Report to the G20, supra* note 8, at 11.

time data replication, with operators meeting C.P.M.I.–IOSCO’s two-hour recovery-time objective for critical infrastructures.<sup>90</sup>

F.P.S.s should also maintain hot-standby connections to correspondent banking channels to reroute traffic during outages. The Single Rulebook should mandate coordinated incident response plans, clear communication protocols, and regular live-fire failover drills to ensure P.S.P.s and merchants can switch rails smoothly. While correspondent banking may be slower or costlier, they remain essential for preserving service continuity and sustaining trust in the system.

### **3. *Fraud Risk***

Fraud risk is heightened in F.P.S. interlinking due to real-time settlement, cross-border complexity, and the difficulty of freezing funds once moved abroad. Common threats include phishing, account takeover, identity theft, and social engineering. The speed of transfers enables fraudsters to withdraw or reroute funds across jurisdictions before detection.<sup>91</sup> Legal fragmentation compounds the challenge, with inconsistent account-freezing powers and varying application of the *bona fide purchaser for value without notice* rule, which, like settlement finality, gives recipients confidence that legitimate payments cannot be unwound.

The Single Rulebook should explicitly enshrine the *bona fide purchaser* principle while balancing fairness to defrauded payers through capped reimbursement or insurance schemes. Fraud mitigation should follow a layered defence model, incorporating robust authentication,

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<sup>90</sup> *Recommendations and Key Features*, *supra* note 74.

<sup>91</sup> *Final Report to the G20*, *supra* note 8, at 12.

transaction monitoring, shared blacklists, real-time alerts, freezing protocols, and coordinated engagement with agencies such as Interpol.

Where fraudulent funds remain in an account, the Single Rulebook should set a standard recovery process: a victim complaint to their P.S.P., followed by investigation, notification to the receiving F.P.S. or P.S.P., and a court or regulatory order in the receiving jurisdiction to freeze the account. Once frozen, the funds may be placed in escrow or returned to the victim, subject to the relevant legal framework. Because interlinking operators typically lack standing to obtain such orders directly, formal cooperation protocols between F.P.S. operators and domestic regulators are essential. Clearly defined procedures, timelines, and role allocations within the Single Rulebook will enhance enforceability, accelerate recovery, and maintain public trust in the arrangement.<sup>92</sup>

#### **4. *Financial Risks***

For lawyers and regulators drafting a Single Rulebook, the priority is managing four interconnected financial risks: credit, liquidity, settlement, and systemic risk. Credit risk arises when a participant cannot meet obligations, eg, if a P.S.P. in one jurisdiction becomes insolvent before paying a debt owed to a P.S.P. in another. Liquidity risk occurs when a participant is solvent but unable to fund payments on time, such as during a sudden foreign currency shortage in a thin weekend corridor, delaying transfers. Settlement risk emerges when one leg of a cross-currency payment settles but the matching leg fails, leaving the transaction incomplete. Systemic risk is the most severe, occurring when the failure of a major P.S.P. cascades across

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<sup>92</sup> *Results of the 2023 C.P.M.I. Cross-Border Payments Monitoring Survey*, *supra* note 7.

multiple P.S.P.s and F.P.S.s, potentially freezing retail and wholesale payments across jurisdictions.

One way to mitigate these risks in an interlinking arrangement is to update all relevant accounts in real-time, crediting the payee's account only after the payer's account and P.S.P.'s account are debited and the payee's P.S.P.'s account is credited. This sequencing ensures funds are irrevocably transferred through every layer before final credit, reducing financial risks and enhancing certainty.

Implementing this approach across borders is challenging for F.P.S.s that rely on deferred settlement, where transactions clear in scheduled batches rather than continuously.<sup>93</sup> Enabling real-time sequencing in such systems could require fundamental architectural changes, often costly and politically unpalatable. Since the goal of interlinking is to leverage, not replace, existing F.P.S. infrastructure, more practical alternatives include prefunding, collateralisation, or loss-sharing arrangements.

A robust interlinking framework would combine three safeguards. First, P.S.P.s should maintain prefunded balances in central-bank money, either with the F.P.S. or in central bank reserve accounts, to eliminate exposure to P.S.P. insolvency. Second, a well-regulated mutual default fund or C.C.P. can absorb losses exceeding an insolvent P.S.P.'s prefunding. Third, settlement finality legislation must ensure transfers become irrevocable once the payee's account is credited.

Together, prefunding, a loss-sharing backstop, and strong legal finality protections substantially reduce credit, liquidity, and settlement risks while preventing a single failure from

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<sup>93</sup> *Final Report to the G20, supra* note 8, at 9.

escalating into systemic disruption.<sup>94</sup> This integrated approach strengthens resilience and stability while building the trust necessary for long-term success.

## **5. *FX Settlement Risk***

Cross-currency F.P.S. interlinking requires an FX transaction whenever payer and payee use different currencies. This involves exchanging<sup>95</sup> one currency for another at an agreed rate. A market maker, which is often a bank with accounts at both central banks, pays out the sold currency from one account and receives the bought currency into another. These are processed as separate domestic transfers, recorded as an outgoing payment and incoming receipt.<sup>96</sup>

This process exposes the P.S.P. or market maker to FX settlement risk, which is commonly known as Herstatt risk, if one leg settles while the other fails. A payment-versus-payment (PvP) mechanism removes this risk by ensuring both legs settle simultaneously. Where PvP is unavailable, time-zone differences or operational delays can create a settlement gap, leaving the delivering party exposed.

The most famous example occurred on June 26, 1974, with the collapse of Bankhaus Herstatt. The German bank received Deutsche Mark payments from counterparties in Frankfurt during the European trading day, but before it could transfer the corresponding US dollars in New York, it was declared insolvent and shut down at 3:30 PM German time. Because the US payment system had not yet opened, the dollar leg never settled, and counterparties suffered

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<sup>94</sup> “[T]he credit risk borne by the payee’s P.S.P. can be managed, for example, through limits (to the aggregate net positions of P.S.P.s), frequent settlement cycles, loss-sharing agreements, collateralisation and/or prefunding arrangements,” *id.* at 11.

<sup>95</sup> “[C]ross-currency payments require an FX conversion at some point. In the correspondent banking model, this conversion is done by one or more correspondent banks along the payments chain. Often it is done by either the first (i.e., the payer’s) bank or the last (i.e., the payee’s) bank in the chain,” *Towards the Holy Grail*, *supra* note 39, at 45–46.

<sup>96</sup> *Id.*

major losses. The root cause was non-synchronous settlement across time zones, which is a vulnerability that persists where PvP is absent.

In such events, counterparties become unsecured creditors of the insolvent bank, often recovering only a fraction of their claims after lengthy proceedings. This underscores the need for PvP or comparable safeguards in F.P.S. interlinking to prevent settlement gaps from becoming principal losses.

## **6. *Payment-versus-Payment (PvP)***

The PvP mechanism was developed to prevent scenarios like the Herstatt crisis by ensuring both currency legs of an FX transaction settle simultaneously. Under PvP, the final transfer of one currency occurs only if the final transfer of the other currency happens at the same time.

Launched in 2002 by major central banks and FX market participants, the C.L.S. system implements PvP for wholesale FX transactions across 18 major currencies. C.L.S. connects directly to participating currencies' R.T.G.S. systems and holds central bank accounts for settlement, acting as an escrow-like intermediary.<sup>97</sup> In a typical C.L.S. transaction, eg, a euro–USD trade between a German and a US bank, each bank funds C.L.S. by transferring the sold currency to C.L.S.'s account at its central bank. C.L.S. releases each currency to the counterparty only once both legs are fully funded, eliminating settlement risk. If one party fails

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<sup>97</sup> “Technically, PvP settlement can be achieved in different ways. It can be done through a direct link between two payment systems or through a common agent in the middle,” Jakub Demski et al., *Growth of the Green Bond Market and Greenhouse Gas Emissions*, B.I.S. Q. REV., Mar. 2020, at 53, 62, [https://www.B.I.S..org/publ/qtrpdf/r\\_qt2003.pdf](https://www.B.I.S..org/publ/qtrpdf/r_qt2003.pdf).



to fund, C.L.S. returns the other's funds, leaving only replacement cost risk<sup>98</sup> from re-entering the market at a different rate.

C.L.S. operates under a contractual framework in which participants settle with C.L.S. rather than with each other. While highly effective in mitigating settlement risk, C.L.S. is not a 24/7 system; its operations are limited to overlapping R.T.G.S. hours, which can delay settlement. Moreover, C.L.S. is designed for wholesale FX, not retail or instant cross-border payments. Broader PvP adoption remains constrained by the operational, technical, and legal complexities of linking multiple currencies and jurisdictions. Although PvP and C.L.S. have significantly reduced risks like those of 1974, limitations such as restricted operating hours and residual replacement cost risk mean settlement risk is mitigated but not eliminated.

## ***7. Managing FX Settlement Risk***

F.P.S. interlinking falls into three categories: single-currency, multi-currency, and cross-currency links, each with different implications for FX and settlement risk. In single-currency systems (eg, euroSIC, SADC), transactions settle in one designated currency, with FX handled bilaterally outside the link. Multi-currency systems (eg, T.I.P.S.) process several currencies in parallel but require P.S.P.s to manage conversions independently, meaning no cross-currency settlement or embedded FX risk. By contrast, cross-currency links (eg, Directo a México, GCC's AFAQ) embed FX conversion within the system, crediting payees in a different currency and thereby introducing settlement risk that operators must mitigate, ideally through PvP.

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<sup>98</sup> Replacement cost risk in the context of FX transactions refers to the potential financial loss a party faces when a counterparty fails to fulfil its obligations, forcing the affected party to enter a new transaction at potentially less favourable market rates.

While P.S.P.s manage FX risks externally in the first two models, unmanaged conversion risk in cross-currency arrangements can propagate across borders. While 75% of F.P.S. currently operate in a single currency, 43% of multicurrency systems plan to enable cross-currency settlement within five years, which indicates a clear trend toward more integrated FX functionalities in interlinked systems.<sup>99</sup>

FX settlement risk in such arrangements can be mitigated by combining prefunded central bank accounts, multilateral netting, and robust legal settlement finality. Each P.S.P. should maintain prefunded balances, either at its domestic central bank or within the F.P.S., in every currency in which it provides FX conversion. These balances must be sufficient to cover obligations during R.T.G.S. closures, such as weekends or public holidays, when incoming cross-border transactions can accumulate.

At defined intervals, the F.P.S. operator nets<sup>100</sup> each P.S.P.'s outgoing and incoming flows, reducing liquidity needs. For example, if Bank A owes Bank B \$100 and Bank B owes Bank A \$50, only \$50 is transferred when the R.T.G.S. system reopens. If a P.S.P.'s net debit position exceeds its prefunded balance, the operator can automatically suspend<sup>101</sup> further payouts until the account is replenished, ensuring no P.S.P. can send more than it has funded. This process can be automated via APIs and smart contract-based rules.<sup>102</sup>

Legal protection is essential. The Single Rulebook and domestic legislation should grant participating P.S.P.s 'super-priority' creditor status in insolvency. For instance, Article 3 of the SFD (EU)<sup>103</sup> ensures that "netting shall be legally enforceable and binding on third

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<sup>99</sup> *Results of the 2023 C.P.M.I. Cross-Border Payments Monitoring Survey*, *supra* note 7.

<sup>100</sup> *A Playbook for ASEAN and the World*, *supra* note 73, at 17–24.

<sup>101</sup> *Id.*

<sup>102</sup> *Towards the Holy Grail*, *supra* note 39, at 45–46.

<sup>103</sup> *Directive 98/26/EC*, *supra* note 20.

parties even in the event of insolvency proceedings against a participant.” This prevents insolvency administrators from unwinding settlement or netting arrangements if a P.S.P. in the F.P.S. interlinking framework fails.

Super-priority ranking ensures that net creditor P.S.P.s repaid first from an insolvent P.S.P.’s prefunded account, ahead of all other claims, and without waiting for lengthy insolvency proceedings. Funds in these accounts must be ring-fenced so they cannot be clawed back by insolvency administrators. Netting and set-off rights should also be explicitly enforceable, allowing a net debtor P.S.P. to pay only the net amount owed rather than the gross value of unsettled transactions.

This structure mirrors the protections of C.L.S. Bank’s PvP model but extends them to cover both real-time and D.N.S. F.P.S. systems, with the ability to operate on a near-24/7 basis. Crucially, settlement finality must be recognised across all participating jurisdictions so that once a payment is executed, it is irrevocable, even if a P.S.P. fails moments later. Without such statutory protection, cross-border payment flows remain legally vulnerable, undermining system integrity.

Dynamic liquidity safeguards, such as net debit caps and top-up triggers, further strengthen resilience. When an R.T.G.S. is closed and additional funding is needed, P.S.P.s can pledge collateral to their central bank or draw on emergency liquidity lines. Adjustments to reserve requirements and standby funding mechanisms may also be necessary to support continuous operation. Together, these measures, including prefunding, netting, settlement finality, super-priority status, and liquidity support, create a ‘synthetic PvP’ framework. By mitigating settlement risk and ensuring enforceable outcomes, this framework advances the

G20 objective of faster, safer, and more transparent cross-border payments, even in environments without real-time PvP.

## **8. *The Point of Settlement Finality***

From a legal standpoint, the key question is: *when are funds available to the end user, and when does cross-border settlement achieve finality between intermediaries?*<sup>104</sup> The C.P.M.I.–IOSCO Principles for Financial Market Infrastructures (P.F.M.I.) apply to systemically important payment systems,<sup>105</sup> with Principle 8 requiring rules to define settlement finality as the moment a transfer becomes irrevocable and unconditional.

For F.P.S. interlinking arrangements, the Single Rulebook should state unequivocally that finality occurs once the payee’s account is credited.<sup>106</sup> From that moment, the credit is irrevocable and cannot be unwound, *even if* the payer’s P.S.P. has not yet settled with the payee’s P.S.P. or later becomes insolvent.

If the payer’s P.S.P. fails before interbank settlement is completed, the payee’s P.S.P. should be reimbursed from the payer’s prefunded central-bank account held by the link operator (whether in the payer’s or the payee’s jurisdiction). This arrangement gives the payee’s P.S.P. a super-priority claim over the insolvent P.S.P.’s other unsecured creditors.

Similarly, if the payee’s P.S.P. enters resolution later that day, its insolvency administrator must not claw back the incoming credit under any zero-hour doctrine. By

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<sup>104</sup> *Recommendations and Key Features*, *supra* note 74, at 22.

<sup>105</sup> The P.F.M.I. defines FMIs as “multilateral systems among participating institutions, including the operator of the system, used for the purposes of clearing, settling, or recording payments, securities, or other financial transactions,” *P.F.M.I.*, *supra* note 1, at 406. Given that FPS are designed to facilitate the clearing and settlement of retail payments, they clearly fall within this definition.

<sup>106</sup> *Recommendations and Key Features*, *supra* note 74, at 22; *Recommendations and Key Features*, *supra* note 74, at 15.

defining legal finality as the moment the payee’s account is credited, the Single Rulebook insulates that entry from reversal once it appears on the payee’s ledger.

The Single Rulebook should also require that the link operator to connect F.P.S. systems only in jurisdictions with enforceable settlement finality legislation—or, where absent, on the basis of a legal opinion confirming that settlement finality is recognised for F.P.S.s. It should further clarify that an intervening insolvency—whether of the payer’s P.S.P., the payee’s P.S.P., or either P.S.P.’s correspondent—must not affect the validity of the credit to the payee’s account or the payee’s P.S.P.’s account.

To reinforce compliance, additional incentives, such as fines or expulsion from the interlinking arrangement, should be available to ensure that all participating jurisdictions adhere to this legal standard.

## **9. *Systemic Risk***

Systemic risk in an F.P.S. interlinking arrangement is smaller in scale, but not in consequence, when compared with wholesale R.T.G.S. systems or derivatives markets. For example, the UK Faster Payments Service handled 5.09 billion retail transfers worth £4.2 trillion in 2024, roughly £11 billion daily across 46 direct participants, with a single-payment cap of £1 million.<sup>107</sup> By contrast, the UK’s high-value CHAPS R.T.G.S. settled £542 billion in a single

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<sup>107</sup> *Fast Payment System*, *supra* note 12.

day alone in January 2022,<sup>108</sup> while the global ‘over-the-counter’ (OTC) FX market turns over about USD \$7.5 trillion daily.<sup>109</sup>

Other retail instant-payment rails show similar patterns. The US RTP network processed 343 million transfers worth US \$246 billion in 2024,<sup>110</sup> while Europe’s T.I.P.S. settled 269.8 million payments in 2023 at an average €641 each, roughly €170 billion in total.<sup>111</sup> Individually, these flows are modest next to wholesale payment systems. But when dozens of domestic F.P.S.s are interconnected, an outage at the interlinking operator, or exhaustion of its default resources, could freeze retail payments across multiple countries and spread to the wider financial system.

Individually, most F.P.S.s would not qualify as systemically important. Yet interlinking them could create systemically significant infrastructure, amplifying risk if a major participant’s failure drains liquidity across the network. Such arrangements should therefore align with the P.F.M.I. to bolster resilience and harmonise oversight, even absent formal systemic designation.<sup>112</sup>

## **10. A C.C.P. for Interlinking Arrangements**

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<sup>108</sup> BANK OF ENGLAND, REAL-TIME GROSS SETTLEMENT (R.T.G.S.) SYSTEM AND CHAPS ANNUAL REPORT 2021/22 (Sep. 6, 2022), <https://www.bankofengland.co.uk/report/2022/R.T.G.S.-and-chaps-annual-report-2021-22>.

<sup>109</sup> OTC Derivatives Statistics, B.I.S. DATA PORTAL, [https://data.B.I.S..org/topics/OTC\\_DER](https://data.B.I.S..org/topics/OTC_DER) (last visited Oct. 10, 2025).

<sup>110</sup> *Skyrocketing RTP® Network Growth Sets Stage for 2025*, THE CLEARING HOUSE (Jan. 8, 2025), [https://www.theclearinghouse.org/payment-systems/Articles/2025/01/RTP\\_2024\\_Year\\_Records\\_01-08-2025](https://www.theclearinghouse.org/payment-systems/Articles/2025/01/RTP_2024_Year_Records_01-08-2025).

<sup>111</sup> EUR. CENT. BANK, TARGET ANNUAL REPORT: 2023, at 20–21 (June 2024), <https://www.ecb.europa.eu/pub/pdf/targetar/ecb.targetar2023.en.pdf>.

<sup>112</sup> *Considerations for Governance and Oversight*, *supra* note 43, at 18.

A novel approach would be to establish a global or regional C.C.P. clearing system as the operator of an interlinked F.P.S. A C.C.P. manages counterparty and settlement risk by novating all payment obligations, acting as buyer to every seller and seller to every buyer. Thus, when P.S.P. A initiates a payment to P.S.P. B and the operator accept it, the obligation is extinguished and replaced with two contracts: P.S.P. A owes the C.C.P., and the C.C.P. owes P.S.P. B. This ensures P.S.P. B is paid even if P.S.P. A defaults.

Payouts would come from prefunded resources, including each P.S.P.'s cash collateral at central banks in participating jurisdictions and a rigorously stress-tested default fund. Domestic settlement finality laws would give such claims super-priority, ensuring enforceability in insolvency.

Comparable frameworks already exist in financial markets: C.C.P.s are subject to robust regulation under regimes like EMIR and *Dodd-Frank*, and no major C.C.P. has defaulted since 1974. The value of cross-border transfers a C.C.P. for F.P.S. would clear and settle would be modest compared to derivatives markets (eg, USD 2½ billion per day in remittances<sup>113</sup> versus USD 7.5 trillion daily in the OTC FX market).<sup>114</sup>

A C.C.P. could further enhance dispute resolution for erroneous but authorised payments. Instead of forcing senders into costly foreign litigation, unresolved claims could be routed through the C.C.P., which, using its standing in multiple jurisdictions or pre-agreed arbitration, would wield far more leverage than individual claimants.

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<sup>113</sup> Dilip Ratha, Sonia Plaza & Eung Ju Kim, *In 2024, Remittance Flows to Low- and Middle-Income Countries Are Expected to Reach \$685 Billion, Larger than FDI and ODA Combined*, WORLD BANK BLOGS (Dec. 18, 2024), <https://blogs.worldbank.org/en/peoplemove/in-2024--remittance-flows-to-low--and-middle-income-countries-ar>.

<sup>114</sup> BANK FOR INT'L SETTLEMENTS, MONETARY AND ECON. DEP'T, OTC FOREIGN EXCHANGE TURNOVER IN APRIL 2022 (Oct. 27, 2022), [https://www.B.I.S..org/statistics/rpfx22\\_fx.pdf](https://www.B.I.S..org/statistics/rpfx22_fx.pdf).

The model, however, carries costs. A cross-border C.C.P. would require a common charter, governance, and supervisory college. P.S.P.s would need to post margin and contribute to a mutualised default fund, tying up capital. A C.C.P. also creates concentration risk: its failure would halt settlement system-wide.

Moreover, novation alone does not ensure PvP, as time-zone mismatches may cause the C.C.P. to release one leg of a transaction before receiving the counter-leg. Prefunded collateral mitigates but does not eliminate this risk, making synchronised settlement essential. If the C.C.P. also acts as FX settlement agent, it must hold central bank accounts and maintain liquidity in all relevant currencies, which is an operationally demanding task.

On balance, a C.C.P. could consolidate prefunding, collateral management, and statutory protections into a single clearing utility, significantly reducing credit and settlement risk. Its design could allow real-time and deferred-net F.P.S.s to interoperate securely across time zones and cut-off times. With payments settled in central bank money and finality legally incontestable, recipients could use funds immediately, confident they cannot be clawed back. The trade-off is whether these benefits justify the costs of building a new C.C.P. global infrastructure.

## **11. *Default Fund***

Where a fully-fledged C.C.P. is politically or financially untenable, an intermediate alternative is a mutual default fund. Each participating P.S.P. would contribute a small, prefunded amount (perhaps as part of the membership fee), which would be scaled to its payment volumes, into a ring-fenced pool functioning like insurance. If an insolvency court freezes P.S.P. A's prefunded balance before it can be released to P.S.P. B, the fund immediately compensates P.S.P. B and pursues recovery from the insolvent estate later. This model spreads loss contractually across



the network without creating a new systemically important entity, keeps setup costs low, and reassures users that legitimate payments will be honoured even when a counterparty fails.

## **12. AML/CFT Risk**

All payment systems face risks of illicit finance, including money laundering, terrorist financing, and sanctions evasion. These AML/CFT risks are heightened in F.P.S. interlinking due to greater jurisdictional diversity, differing data-sharing rules, and real-time execution. In such environments, reliably identifying both the originator and beneficiary becomes harder, particularly when participants follow divergent messaging standards or when inconsistent legal regimes block information exchange.<sup>115</sup>

Sanctions screening must occur for every transfer, even when the customer has passed standard know-your-customer (K.Y.C.) checks. Sanctions lists (eg, Office of Foreign Assets Control, United Nations, EU) are frequently updated; a customer cleared at onboarding may later appear on a list.<sup>116</sup> Regulators and P.S.P.s must therefore screen all transfers in real-time, regardless of prior risk assessments. Thus, this requirement can delay cross-border transactions.

### *A. Streamlining Compliance*

Given the speed of F.P.S., AML/CFT compliance must be straight-through processable (S.T.P.). This requires automating checks and sharing results across P.S.P.s to minimise redundancy. In theory, if the payer P.S.P. performs sanctions and AML screening at initiation

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<sup>115</sup> *Exploring Multilateral Platforms*, *supra* note 30, at 11–13.

<sup>116</sup> *A Playbook for ASEAN and the World*, *supra* note 73.

and the payee P.S.P. trusts the quality of that check, perhaps because both use the same compliance software or screening engines, a second check may be unnecessary.

A uniform compliance model, where results are shared between P.S.P.s across borders, would significantly reduce friction and duplication.<sup>117</sup> P.S.P.s could secure customer consent during onboarding to allow cross-jurisdictional information sharing strictly for AML/CFT compliance.

Project Nexus and the B.I.S. Innovation Hub have explored embedding screening APIs directly into the interlinking hub to harmonise screening across jurisdictions and reduce duplicate checks.<sup>118</sup> However, such shared infrastructure requires harmonised data protection laws and clear liability rules, particularly if a P.S.P. misses a red flag and illicit funds enter the system.

### *B. International Oversight*

International coordination is essential. All F.P.S.s seeking to join the interlink should comply with the FATF Recommendations, with jurisdictions on the FATF greylist or blacklist barred or automatically suspended to mitigate financial crime risks. This creates strong incentives for jurisdictions to adopt and enforce robust AML/CFT controls, reinforced by FATF's periodic evaluations of participating F.P.S.s.

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<sup>117</sup> *Towards the Holy Grail*, *supra* note 39, at 42–43.

<sup>118</sup> *Exploring Multilateral Platforms*, *supra* note 30, at 5–25; *Interlinking Fast Payment Systems*, *supra* note 3.

The link operator must play an active oversight role<sup>119</sup> by hosting a shared compliance scheme, monitoring anomalies across the transaction chain, and enforcing uniform technical standards for sanctions screening, K.Y.C. data, and error handling. The Single Rulebook should formalise these responsibilities and grant the operator clear enforcement powers, including suspension or disconnection of non-compliant P.S.P.s or F.P.S.s. Automatic suspension should also apply if a jurisdiction is placed on the FATF greylist or blacklist or if a P.S.P. fails to meet minimum compliance thresholds.

Given the speed of cross-border F.P.S., law enforcement coordination must be equally agile. Interpol can serve as a conduit between Financial Intelligence Units and national authorities, helping freeze or recover illicit funds when flagged transactions move across jurisdictions.

### *C. Example of Good Practice*

The Singapore–Thailand PayNow–PromptPay linkage offers a practical case study in coordinated compliance. Both countries adopted real-time screening engines and imposed transaction caps (SGD 1,000 or THB 25,000) to mitigate the risk of funds being paid out before screening was completed. Suspicious transfers were rejected before inclusion in the daily settlement file, preventing illicit funds from entering the financial system.<sup>120</sup>

While both countries adhered to FATF standards, they applied compliance differently: Singapore conducted AML/CFT screening at the transfer phase—just before funds are

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<sup>119</sup> *Exploring Multilateral Platforms*, *supra* note 30, at 5–25. Panicos Demetriades & Radosveta Vassileva, *Money Laundering and Central Bank Governance in the European Union*, 23 J. INT’L ECON. L. 509, 512 (2020) (“central banks usually have an important role in the fight against money laundering”).

<sup>120</sup> *A Playbook for ASEAN and the World*, *supra* note 73, at 17–24.

debited—whereas Thailand screened earlier, during the lookup phase, when the payer searched for the payee’s name in the online banking interface. The linkage succeeded because both sides agreed on a minimum dataset, harmonised workflows, and a shared understanding of when and how screening would occur.

#### *D. Design Considerations for Interlinking Arrangements*

Beyond real-time screening and international coordination, future interlinking arrangements may adopt tiered, risk-based models. For example, transactions exceeding USD 10,000 might be routed into a delayed processing queue for additional scrutiny, with exceptions for pre-screened corporate clients with strong compliance records. Countries could also agree on minimum transaction limits or let users set personal caps, particularly for high-risk corridors prone to fraud or ML/TF.

Regulatory alignment and technological innovation can reduce AML/CFT risk without undermining the speed of fast payments. The B.I.S.’s Project Mandala, for instance, explores automating the policy compliance layer, including sanctions, AML/CFT, and capital control, before a cross-border payment is initiated.<sup>121</sup> As regulators and link operators expand supervisory scope, these embedded safeguards can help ensure that F.P.S. interlinking remains both efficient and secure.

Furthermore, liability for AML/CFT failures must be clearly defined in the Single Rulebook. Typically, each P.S.P. remains responsible for its jurisdiction’s compliance obligations, including screening and monitoring. However, if illicit transactions are processed due to negligence, such as a failure to screen a sanctioned party or complete due diligence, the

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<sup>121</sup> *Interlinking Fast Payment Systems*, *supra* note 3.

P.S.P. at fault may be held liable by its regulator or other participants. The Single Rulebook should therefore specify provisions for dispute resolution, remediation, and cost allocation in such cases.

### **13. *Data Protection and Privacy***

Protecting consumer data is essential in any F.P.S. interlinking arrangement. Given the sensitivity of financial and personal data exchanged across jurisdictions, participants must adopt robust legal, contractual, and technical safeguards to comply with local data protection laws and international best practices.

At a broader level, inconsistent data privacy laws across jurisdictions remain a challenge for F.P.S. interlinking. Some countries require in-country processing and storing (data localisation), which can hinder features like pre-validation or confirmation-of-payee. The use of proxy identifiers, such as phone numbers or emails, also raises concerns, as these are often considered personally identifiable information (PII).

The Singapore–Thailand PayNow–PromptPay linkage illustrates good practice. Its Multi-Party Agreement permits banks to collect and process customer data solely to enable transfers. This contractual restriction aligns with national privacy laws in both jurisdictions and required banks to update their terms and conditions, privacy notices, and customer agreements.<sup>122</sup> Customers were then given the choice to accept the updated terms, reinforcing transparency and consent as core principles.

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<sup>122</sup> *A Playbook for ASEAN and the World*, *supra* note 73, at 17–24.

To prevent misuse of personal data, such as name farming during proxy lookups, technical solutions were implemented. For example, when a sender in Singapore performs a proxy lookup using a recipient's mobile number, only a masked version of the recipient's name appears (eg, 'TH\*\*\*\* LE\*'), preserving anonymity while still confirming the correct payee. Both countries also enabled opt-out functionality: Thai users could receive PromptPay transfers while opting out of cross-border ones, with similar enhancements underway in Singapore. Additional controls included rate limits on proxy queries to prevent abuse.

Harmonising national data privacy standards, through frameworks like the Global Cross-Border Privacy Rules Forum, can ease compliance burdens and enable secure cross-border data-sharing.<sup>123</sup> Ultimately, the interlinking arrangement's Single Rulebook should clearly define privacy obligations, designate responsible parties, and mandate protections such as masking, opt-outs, and rate-limiting to strengthen user trust and legal certainty.

## VIII SELECTED PROJECTS

Project Nexus is a flagship initiative led by the B.I.S. Innovation Hub with the central banks of Singapore, Malaysia, Indonesia, the Philippines, and Thailand. Nexus aims to interlink F.P.S. via a common API-based architecture and a centralised governance model. Instead of deep harmonisation, it relies on a 'service level description' for each country, codifying local rules and formats (eg, alias usage, timeouts, dispute periods). This allows payments to reach their destination in under 60 seconds, even across multiple currencies.<sup>124</sup>

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<sup>123</sup> *Interlinking Fast Payment Systems*, *supra* note 3; *Global CBPR Forum: Building Digital Trust through Partnerships*, GLOBAL CBPR FORUM (2024), <https://www.globalcbpr.org/>.

<sup>124</sup> *P.F.M.I.*, *supra* note 1.

Another ambitious initiative is the TARGET Instant Payment Settlement (T.I.P.S.) system in Europe. T.I.P.S. enables instant settlement in central bank money across the euro area and is evolving into a multicurrency platform. Sweden's RIX-INST is already integrated, allowing real-time cross-currency payments between the euro and Swedish krona. T.I.P.S. also supports multiple Automated Clearing Houses and enables P.S.P.s to shift liquidity across systems on a 24/7 basis, including weekends. Denmark's kroner is next in line for integration, using cross-currency technical accounts held by central banks to facilitate settlement.<sup>125</sup>

Project Rialto, launched in 2024 by the B.I.S. Innovation Hub, goes further by addressing FX and settlement frictions in retail cross-border payments. Its proof of concept leverages an automated wholesale FX conversion layer in central bank money, eliminating commercial bank risk from transactions. Participants include the Bank of France, Bank of Italy, MAS, and Bank Negara Malaysia. Rialto could significantly reduce remittance costs.<sup>126</sup>

The Amplus concept, proposed by the Deutsche Bundesbank, targets remittance corridors specifically. It envisions central banks participating directly in a global multicurrency system, with domestic P.S.P.s accessing it indirectly. The model integrates a K.Y.C. identifier and addressability framework to streamline compliance and expand access, especially for underbanked populations.

Together, these initiatives show the diversity interlinking models,<sup>127</sup> allowing jurisdictions to adopt frameworks suited to their technical, legal, and policy contexts. At the

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<sup>125</sup> BANK FOR INT'L SETTLEMENTS, COMM. ON PAYMENTS AND MARKET INFRASTRUCTURES, INTERLINKING PAYMENT SYSTEMS AND THE ROLE OF APPLICATION PROGRAMMING INTERFACES: A FRAMEWORK FOR CROSS-BORDER PAYMENTS (July 2022), <https://www.B.I.S..org/C.P.M.I./publ/d205.pdf>; *Call for Expression of Interest in Supporting the Implementation of the T.I.P.S. Cross-Currency Service*, EUR. CENT. BANK ( Jan. 17, 2025), [https://www.ecb.europa.eu/press/intro/news/html/ecb.mipnews250117\\_1.en.html](https://www.ecb.europa.eu/press/intro/news/html/ecb.mipnews250117_1.en.html).

<sup>126</sup> *P.F.M.I.*, *supra* note 1.

<sup>127</sup> *Consolidated Progress Report for 2024*, *supra* note 45.

same time, they provide valuable lessons that can inform the development of a Single Rulebook or the design of a new multilateral platform to interconnect existing F.P.S.

## **IX CONCLUSION AND RECOMMENDATIONS**

This article has presented the first comprehensive legal and policy framework for the safe and effectively interlinking of F.P.S. across borders. While recent initiatives have focused largely on technical pilots, this paper highlights the equally essential legal foundations needed to build trust, minimise risk, and achieve the G20's 2027 target of reducing remittance costs below three per cent.

A central recommendation is the establishment of a consistent, enforceable framework across jurisdictions, supported by a Single Rulebook that defines settlement finality, liability, AML/CFT compliance, and dispute resolution. Settlement finality should occur when the payee's account is credited, ensuring legal certainty and protection against reversal. Risk management mechanisms, such as prefunded accounts, default funds, or C.C.P. models, must be paired with robust governance, enforceable sanctions, and clear legal opinions confirming enforceability. FATF high-risk jurisdictions should be excluded until deficiencies are resolved.

Unlike wCBDCs or stablecoins, F.P.S. interlinking leverages existing infrastructure, offering a faster and more practical path to cheaper, quicker, and more inclusive cross-border payments. Success will require international cooperation, led by central banks and supported by global institutions such as the B.I.S., C.P.M.I., F.S.B., I.M.F., and World Bank. These bodies



should define common standards, provide technical assistance, and ensure policy alignment, drawing lessons from C.L.S., SWIFT, and G20-led post-2008 OTC clearing reforms.

Regional hubs are likely to emerge first, serving as testbeds for harmonisation before scaling into a global network. But political coordination remains the greatest challenge: aligning domestic laws on settlement finality, netting, data protection, and AML/CFT will be essential to avoid fragmentation.

Ultimately, the interlinking of F.P.S. is an international economic and public-policy imperative. Like central clearing for OTC derivatives, it will demand legal certainty, enforceable rulebooks, and sustained international cooperation. The reward is faster, safer, and more accessible cross-border payments for billions of people worldwide.